HW0505

11.24 Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

a. Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for ln(*PRICE*), adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?

b. Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.

c. Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.

d. In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

a.

```
> rf_model <- lm(lprice ~ stormy + MIXED + mon + tue + wed + thu, data = fultonfish)
> summary(rf_model)

Call:
lm(formula = lprice ~ stormy + MIXED + mon + tue + wed + thu,
    data = fultonfish)

Residuals:
     Min       1Q   Median       3Q      Max
-0.92281 -0.18819  0.00813  0.22404  0.68381

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.35957    0.07915  -4.543 1.50e-05 ***
stormy       0.44634    0.07921   5.635 1.51e-07 ***
MIXED        0.23688    0.07851   3.017  0.00321 **
mon         -0.10825    0.10339  -1.047  0.29753
tue         -0.06609    0.10103  -0.654  0.51446
wed         -0.04927    0.10377  -0.475  0.63591
thu          0.03935    0.10071   0.391  0.69681
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3413 on 104 degrees of freedom
Multiple R-squared:  0.245,	Adjusted R-squared:  0.2014
F-statistic: 5.624 on 6 and 104 DF,  p-value: 4.349e-05
```

p-value = 0.00321 < 5% ⇒ statistically significant.

```
> summary(rf_model)$coefficients["MIXED", ]
   Estimate  Std. Error     t value    Pr(>|t|)
0.236875284 0.078513425 3.017003561 0.003209729
> joint_test <- linearHypothesis(rf_model, c("stormy = 0","MIXED = 0"))
> joint_test

Linear hypothesis test:
stormy = 0
MIXED = 0

Model 1: restricted model
Model 2: lprice ~ stormy + MIXED + mon + tue + wed + thu

  Res.Df    RSS Df Sum of Sq     F    Pr(>F)
1    106 15.876
2    104 12.115  2    3.7604 16.14 7.857e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> F_value <- joint_test$F[2]
> F_value
[1] 16.14003
```

> 10.

b.

```
> iv_model <- ivreg(lquan ~ lprice + mon + tue + wed + thu |stormy + MIXED + mon + tue + wed + thu,
data = fultonfish)
> summary(iv_model)

Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | stormy +
    MIXED + mon + tue + wed + thu, data = fultonfish)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1327 -0.4320  0.0654  0.5133  1.1798

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533  < 2e-16 ***
lprice      -0.93014    0.35340  -2.632  0.00977 **
mon         -0.01190    0.20837  -0.057  0.95456
tue         -0.52583    0.20230  -2.599  0.01068 *
wed         -0.56262    0.20696  -2.719  0.00767 **
thu          0.09987    0.20285   0.492  0.62350
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared: 0.185,	Adjusted R-squared: 0.1462
Wald test: 4.927 on 5 and 105 DF,  p-value: 0.0004317
```

lprice的係數為-0.93014，小於table 11.5中的-1.1194，但仍顯著.

c.

```
> iv_model_diag <- summary(iv_model,diagnostics = TRUE)
> iv_model_diag$diagnostics
                df1 df2  statistic      p-value
Weak instruments  2 104 16.1400276 7.856950e-07
Wu-Hausman        1 104  1.4891272 2.251117e-01
Sargan            1  NA  0.7721695 3.795467e-01
> sargan_test <- iv_model_diag$diagnostics["Sargan", ]
> sargan_test
        df1       df2 statistic   p-value
1.0000000        NA 0.7721695 0.3795467
```

H_0: instruments are valid

Sargan-test p-value = 0.3795467 > 5%

⇒ H_0 isn't rejected, and the instruments are valid.

d.

```
> weekday_test <- linearHypothesis(rf_model, c("mon = 0","tue = 0","wed = 0","thu = 0"))
> weekday_test

Linear hypothesis test:
mon = 0
tue = 0
wed = 0
thu = 0

Model 1: restricted model
Model 2: lprice ~ stormy + MIXED + mon + tue + wed + thu

  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    108 12.408
2    104 12.115  4   0.29256 0.6279 0.6437
```

F = 0.6279 < 10. p-value = 0.6437 > 5%

⇒ H_0 isn't rejected, and the variables MON, TUE, WED, THU are not jointly significant in the reduce-form equation, so they don't explain price well.

⇒ Not able to be confident to estimate the supply equation by 2SLS.

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

a. Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?

b. Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?

c. Estimate the price elasticity of demand "at the means" using the results from (b).

d. Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI, PS, and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.

e. What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?

f. Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

a.

Demand: $Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}$

$$\rightarrow P_i = -\frac{\alpha_1}{\alpha_2} + \frac{1}{\alpha_2}Q_i - \frac{\alpha_3}{\alpha_2}PS_i - \frac{\alpha_4}{\alpha_2}DI_i - \frac{1}{\alpha_2}e_{di}$$

$$= r_1 + r_2 Q_i + r_3 PS_i + r_4 DI_i + u$$

$r_2 = \frac{1}{\alpha_2} < 0$, $r_3 = -\frac{\alpha_3}{\alpha_2} > 0$, $r_4 = -\frac{\alpha_4}{\alpha_2} > 0$.

Supply: $Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$

$$\rightarrow P_i = \frac{1}{\beta_2}Q_i - \frac{\beta_1}{\beta_2} - \frac{\beta_3}{\beta_2}PF_i - \frac{1}{\beta_2}e_{si}$$

$$= \delta_1 + \delta_2 Q_i + \delta_3 PF_i + u_{si}$$

$\delta_2 = \frac{1}{\beta_2} > 0$, $\delta_3 = -\frac{\beta_3}{\beta_2} > 0$.

b.

```
> demand_iv <- ivreg(p ~ q+ps+di | di+pf+ps, data = truffles)
> summary(demand_iv)

Call:
ivreg(formula = p ~ q + ps + di | di + pf + ps, data = truffles)

Residuals:
    Min      1Q  Median      3Q     Max
-39.661  -6.781   2.410   8.320  20.251

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -11.428     13.592  -0.841  0.40810
q             -2.671      1.175  -2.273  0.03154 *
ps             3.461      1.116   3.103  0.00458 **
di            13.390      2.747   4.875 4.68e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-Squared: 0.5567,     Adjusted R-squared: 0.5056
Wald test: 17.37 on 3 and 26 DF,  p-value: 2.137e-06
> supply_iv <- ivreg(p ~ q+pf | di+pf+ps, data = truffles)
> summary(supply_iv)

Call:
ivreg(formula = p ~ q + pf | di + pf + ps, data = truffles)

Residuals:
    Min      1Q  Median      3Q     Max
-9.7983 -2.3440 -0.6281  2.4350 11.1600

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -58.7982     5.8592  -10.04 1.32e-10 ***
q             2.9367     0.2158   13.61 1.32e-13 ***
pf            2.9585     0.1560   18.97  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-Squared: 0.9486,     Adjusted R-squared: 0.9448
Wald test: 232.7 on 2 and 27 DF,  p-value: < 2.2e-16
```

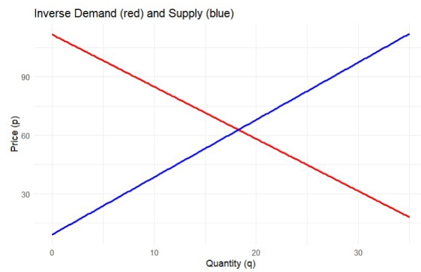
Signs are all correct. 且所有的 p-value 都在至少5%的水準下顯著.

$\Rightarrow$ strong statistical significant

c.

```
> delta1 <- coef(demand_iv)["q"]
> Pbar <- mean(truffles$p)
> Qbar <- mean(truffles$q)
> elasticity <- (1 / delta1) * (Pbar / Qbar)
> elasticity
        q
-1.272464
```

d.



e.

```
> delta0 <- coef(demand_iv)["(Intercept)"] + coef(demand_iv)["ps"]*PS0 + coef(demand_iv)["di"]*DI0
> gamma0 <- coef(supply_iv)["(Intercept)"] + coef(supply_iv)["pf"]*PF0
> delta1 <- coef(demand_iv)["q"]
> gamma1 <- coef(supply_iv)["q"]
>
> q_star <- (delta0 - gamma0) / (gamma1 - delta1)
> p_star <- invDem(q_star)          # or invSup(q_star)
>
> c(p_star = round(p_star, 2), q_star = round(q_star, 2))
p_star.(Intercept) q_star.(Intercept)
             62.84              18.25
```

predicted equilibrium: P: 62.84, Q: 18.25

兩種方式高度一致.

f.

```
> supply_ols <- lm(p ~ q+pf+ps,data = truffles)
> summary(supply_ols)

Call:
lm(formula = p ~ q + pf + ps, data = truffles)

Residuals:
   Min     1Q Median     3Q    Max
-6.641 -2.685  0.080  1.920 10.817

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -54.2156     5.1465 -10.534 7.09e-11 ***
q             2.4411     0.2618   9.325 8.91e-10 ***
pf            2.8377     0.1659  17.109 1.14e-15 ***
ps            0.3321     0.2996   1.108    0.278
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.185 on 26 degrees of freedom
Multiple R-squared:  0.9552,    Adjusted R-squared:   0.95
F-statistic: 184.9 on 3 and 26 DF,  p-value: < 2.2e-16
> demand_ols <- lm(p ~ q+di,data = truffles)
> summary(demand_ols)

Call:
lm(formula = p ~ q + di, data = truffles)

Residuals:
     Min       1Q   Median       3Q      Max
-18.1557  -4.8963   0.6921   4.6754  22.8667

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1.6470     7.9975  -0.206   0.8384
q             0.8654     0.4188   2.066   0.0485 *
di           13.7221     1.8564   7.392 5.96e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.482 on 27 degrees of freedom
Multiple R-squared:  0.7612,    Adjusted R-squared:  0.7435
F-statistic: 43.03 on 2 and 27 DF,  p-value: 4.012e-09
```

sign 除了 demand 的之外，其他的都正確

係數除了 demand 的截距和q之外，其他的都更顯著.